

MA 214 Week 12: Lab 4

Conjugate Families: Beta-Binomial

1. **Your name:** _____

BU ID: _____

2. **Partner's name:** _____

BU ID: _____

Big idea: A Beta prior and Binomial likelihood form a conjugate pair, so updating beliefs about a coin's probability of heads only requires adding heads and tails. **Do not use GenAI tools for this lab.** The point is to practice your own analysis work, coding skills, and statistical reasoning.

Files used: coin files in Lab4_data/, lab4_post_data.csv, and lab-starter.R

Skills practiced: Beta priors, conjugate posterior updates, `qbeta()`, sequential updating, and credible intervals.

Your mission: Estimate an unknown coin probability π and describe uncertainty using a Beta-Binomial posterior.

Icebreaker

Before opening R: introduce yourself to your partner and answer the warm-up question below.

Warm-up response: If you had no coin data yet, what Beta prior would you choose for the probability of heads? Why?

Getting Started

Open lab-starter.R. Choose or use your assigned Lab 4 coin file and compute the number of heads and tails.

```
library(dplyr)
library(ggplot2)

my_coin_file <- file.path("Lab4_data", "lab4_coin_01.csv")
coin_df <- read.csv(my_coin_file)

n_flips <- nrow(coin_df)
n_heads <- sum(coin_df$result)
n_tails <- n_flips - n_heads
```

HUDDLE UP

Before calculating anything, what does the sample proportion suggest about π ? How much should your prior influence the answer?

Question 1. Explore the Coin Data

Record the observed number of heads and tails, then choose prior parameters.

```
n_heads
n_tails
n_heads / n_flips

alpha_prior <- _____
beta_prior <- _____
```

Variable or data set	Type or role	What it means
alpha_prior		
beta_prior		
n heads		

Write 2-3 sentences explaining your prior choice and what the raw data suggest.

Question 2. Make a Prediction

Before computing the posterior, predict whether the posterior mean will be above, below, or close to 0.50.

PREDICTION TIME

Use your prior and the sample proportion to predict where the posterior mean will land.

```
alpha_post <- alpha_prior + n_heads
beta_post <- beta_prior + n_tails

post_mean <- alpha_post / (alpha_post + beta_post)
cred_int <- qbeta(c(0.025, 0.975), alpha_post, beta_post)
```

What did your team predict? What evidence did you check first?

Question 3. Compute Sequential Updates

Update the Beta parameters one flip at a time and compare with the batch posterior.

```
a_seq <- rep(NA, n_flips + 1)
b_seq <- rep(NA, n_flips + 1)
a_seq[1] <- alpha_prior
b_seq[1] <- beta_prior

for (i in 1:n_flips) {
  a_seq[i + 1] <- a_seq[i] + coin_df$result[i]
  b_seq[i + 1] <- b_seq[i] + (1 - coin_df$result[i])
}
```

HUDDLE UP

Does sequential updating lead to the same final α and β as the all-at-once update? Why?

CLAIM, EVIDENCE, CONTEXT

The posterior estimate of π is _____.

My evidence is _____.

This matters because _____.

Question 4. Interpret in Context

Use the posterior mean and credible interval to describe the coin.

Question	Your answer
What comparison did you make?	
What result is strongest?	
What limitation or caution should readers know?	

DEBATE CORNER

If you and your partner disagree about how strong the evidence is, write both arguments first. Then decide what claim the posterior actually supports.

Question 5. Close the Lab

Submit your completed `lab-starter.R`. Your script should define the posterior summaries requested by your instructor.

Final response: state your estimate of π , cite a credible interval, and explain the context.

Optional Challenge

Plot the posterior density and mark the posterior mean and credible interval.

```
# Optional challenge starter code.  
curve(dbeta(x, alpha_post, beta_post), from = 0, to = 1)
```